

Final QA Portfolio

DevelopersHub Corporation — Software Quality Assurance Program

To-Do List & Login/Signup Web Application

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1. Introduction & Objective

This document consolidates all Quality Assurance work completed across a 6-week SQA training program at DevelopersHub Corporation, testing a To-Do List & Login/Signup web application. The program progressed from manual functional testing through black-box techniques, cross-browser compatibility, Selenium/API automation, performance/load testing, security testing, and finally CI/CD pipeline integration. This portfolio presents an end-to-end, professional view of QA practice suitable for job applications and interviews.

2. Week 1 — Manual Functional & UI Testing

Testing began with 18 manually executed test cases covering the Login, Signup, To-Do, and UI modules, following the Software Testing Life Cycle (STLC). A structured test case register was maintained with preconditions, steps, expected vs. actual results, and pass/fail status.

Metric	Result
Total Test Cases	18
Passed	9
Failed	9
Pass Rate	50%
Total Bugs Logged	9

Sample Test Cases

Test ID	Module	Description	Status
TC-001	Login	Valid login with correct credentials	Pass
TC-003	Login	Login with empty email and password fields	Fail
TC-007	Signup	Password and confirm password mismatch	Fail
TC-010	To-Do	Add a task with empty title	Fail
TC-015	UI	Responsive layout on mobile viewport (375px)	Fail

Bug Severity Breakdown

Severity	Count
Critical	1

High	4
Medium	3
Low	1

Key bugs: empty login form has no validation (BUG-001); password mismatch not caught on signup (BUG-002); invalid email causes HTTP 500 error (BUG-003); login form overflows on 375px mobile viewport (BUG-007).

Full details in week-1/SQA_Week1_TestCaseDocument.docx and SQA_Week1_BugReport.xlsx

3. Week 2 — Black-Box Techniques, Compatibility & Retesting

Week 2 applied structured black-box testing techniques — Equivalence Partitioning (EP) for username input validation and Boundary Value Analysis (BVA) for age-field boundaries (17, 18, 19 ... 64, 65, 66). Cross-browser compatibility testing was performed across Chrome, Firefox, and Edge, and all Week 1 bugs were retested.

Metric	Result
Total Test Cases (Week 2)	57
Passed	51 (89%)
Failed / Partial	6 (11%)
Week 1 Bugs Retested	10 (8 closed, 1 reopened, 1 deferred)

Cross-Browser Compatibility Highlights

Feature	Chrome	Firefox	Edge
Login / Registration	Pass	Pass	Pass
Dropdown Menus	Pass	Partial	Pass
Date Picker	Pass	Pass	Partial
Modal Dialogs	Pass	Partial	Pass
Responsive Layout (1366px)	Pass	Pass	Pass

Browsers tested: Chrome 124, Firefox 126, Edge 124 on Windows 11. 5 browser-specific defects were identified (e.g. Firefox dropdown closing unexpectedly, Edge date-picker alignment).

Full details in week-2/Compatibility_Retesting_Report_Week2.docx, black box testing.docx, and Test_Plan_Document_Week2.docx

4. Week 3 — Automation & API Testing

Week 3 introduced test automation. Selenium WebDriver (Python) automated the two highest-priority flows — Login Validation and To-Do Dashboard functionality — using the Page Object Model (POM) for maintainability, with pytest as the test runner. Postman was used to validate the application's REST API across GET, POST, PUT, and DELETE methods.

Metric	Result
Test Cases Automated	3 (Selenium)
API Methods Tested	GET, POST, PUT, DELETE (Postman)
Automation Framework	Selenium WebDriver + pytest, Page Object Model
Cumulative Bugs Logged (Wk 1–3)	14

Full details in week-3/Automation scripts.docx and Week3_Final_QA_Project_Report.pdf

5. Week 4 — Performance / Load Testing

Load testing was performed using Apache JMeter against the application's Login and Dashboard flows (OrangeHRM Demo), with 50 virtual users. Results were captured via Summary Report, Graph Results, and Response Time Graph listeners.

Metric	Result
Virtual Users	50
Samples	280
Average Response Time	24,638 ms
Max Response Time	65,363 ms
Error Rate	50.71%
Throughput	2.7 requests/min

The high error rate and elevated response times were traced to an overly aggressive 1-second ramp-up period for 50 users; the Test Plan recommends re-running with a 60–100 second ramp-up to obtain a cleaner performance baseline. See Recommendations in the full report.

Full details in performance/Performance_Test_Plan.docx, Performance_Testing_Report.pdf, and the JMeter test script Saad-Login.jmx

6. Week 5 — Security Testing (OWASP Top 10)

Security testing was performed based on the OWASP Top 10, using OWASP ZAP / Burp Suite Community Edition for passive scanning, combined with manual tests for XSS, SQL Injection,

authentication, and session management weaknesses.

Vulnerability	Severity	Status
Reflected XSS	High	Identified
SQL Injection	Critical	Identified
Weak Session Timeout	Medium	Identified
Missing Security Headers	Low	Identified

Full details in security/Vulnerability_Report.docx and Security_Testing_Summary.pdf

7. Week 6 — CI/CD Integration

Automated Selenium tests were designed to integrate into a GitHub Actions pipeline, triggered on every push, pull request, and a nightly schedule. Test results are generated as HTML/JUnit reports and published as pipeline artifacts, with a failing test suite acting as a quality gate that blocks deployment.

Full workflow design and sample YAML pipeline in ci-cd/CICD_Workflow.docx

8. Tools & Technologies Used

Tool / Technology	Purpose	Week Used
Selenium WebDriver (Python)	UI Automation Testing	Week 3, 6
Postman	REST API Testing	Week 3
pytest	Python Test Runner & Reporting	Week 3, 6
Chrome / Firefox / Edge	Cross-Browser Compatibility	Week 2
Chrome DevTools	Mobile Emulation, Network Timing	Week 1–2
Apache JMeter	Performance / Load Testing	Week 4
OWASP ZAP / Burp Suite CE	Security Testing	Week 5
Git / GitHub / GitHub Actions	Version Control & CI/CD	Week 6
Excel / Word / PDF	Test Case & Bug Documentation	Week 1–6

9. Skills Demonstrated

- Manual functional and UI test case design and execution (STLC)
- Black-box testing techniques — Equivalence Partitioning, Boundary Value Analysis

- Cross-browser compatibility testing and defect retesting
- Defect/bug reporting, severity classification, and tracking
- Selenium WebDriver automation using the Page Object Model
- REST API testing with Postman (GET, POST, PUT, DELETE)
- Performance and load testing with Apache JMeter
- Security testing aligned with OWASP Top 10
- Git/GitHub version control for QA artifacts
- CI/CD pipeline design for automated testing (GitHub Actions)
- Professional QA documentation and reporting

10. Conclusion

This portfolio demonstrates a complete, professional QA skill set built over a 6-week structured program — spanning manual testing, black-box techniques, cross-browser compatibility, automation, API testing, non-functional testing (performance and security), and DevOps integration. All supporting artifacts are version-controlled in the accompanying GitHub repository, reflecting real-world QA workflow practices end-to-end.